

Assignment

```
import java.util.Scanner
```

```
public class assignment {
```

```
    public static void main(String[] args) {  
        UserInterface ui = new UserInterface();  
        ui.mainMenu();  
    }
```

```
}
```

```
class Account {
```

```
    private int accountNumber;
```

```
    private String accountHolder;
```

```
    private double balance;
```

```
    private String email;
```

```
    private String phoneNumber;
```

```
    public Account(int accountNumber, String accountHolder, double  
        balance, String email, String phoneNumber) {
```

```
        this.accountNumber = accountNumber;
```

```
        this.accountHolderName = accountHolderName;
```

```
        this.balance = balance;
```

```
        this.email = email;
```

```
        this.phoneNumber = phoneNumber;
```

```
}
```

```
    public int getAccountNumber() {  
        return accountNumber;
```

```
}
```



```

public void deposit (double amount) {
    if (amount > 0) {
        balance += amount;
        System.out.println("Deposit Successful. New Balance: " + balance);
    } else {
        System.out.println("Invalid deposit amount");
    }
}

```

```

public void withdraw (double amount) {
    if (amount > 0 && amount <= balance) {
        balance -= amount;
        System.out.println("Withdraw Successful. New balance: " + balance);
    } else if (amount > balance) {
        System.out.println("Insufficient balance");
    } else {
        System.out.println("Invalid withdrawal amount");
    }
}

```

```

public void displayAccountDetails () {
    System.out.println("Account Number" + accountNumber);
    System.out.println("Account Holder" + accountHolderName);
    System.out.println("Balance" + balance);
    System.out.println("Email" + email);
    System.out.println("Phone Number" + phoneNumber);
}

```



```

public void updateContactDetails (String email, String phonenumber) {
    this.email = email;
    this.phonenumber = phonenumber;
    System.out.println ("contact details updated successfully");
}

```

```

}
class UserInterface {
    private Account[] accounts;
    private int accountCount;
    private Scanner scanner;
    private final int MaxAccounts = 100;

```

```

public UserInterface () {
    accounts = new Account [MaxAccounts];
    accountCount = 0;
    scanner = new Scanner (System.in);
}

```

```

public void createAccount () {
    if (accountCount >= MaxAccounts) {
        System.out.println ("cannot create more account");
        return;
    }

```

```

    System.out.println ("Enter Account Number :");
    int accNum = scanner.nextInt();
    scanner.nextLine();
    System.out.println ("Enter Account Holder name :");
    String name = scanner.nextLine();
    System.out.println ("Enter Initial Balance :");
    double balance = scanner.nextDouble();

```



```

double Scanner.nextLine();
System.out.println("Enter Email");
String email = Scanner.nextLine();
System.out.println("Enter phone Number");
String phone = Scanner.nextLine();
account[accountCount++] = new Account(accountNum, name, balance, email, phone);
System.out.println("Account created Successfully");
}

```

```

private Account findAccount(int accountNum) {
    for (int i = 0; i < accountCount; i++) {
        if (account[i].getAccountNumber() == accountNum)
            return account[i];
    }
    return null;
}

```

```

}
public void performDeposit() {
    System.out.println("Enter Account Number:");
    int accountNum = Scanner.nextInt();
    System.out.println("Enter Amount to Deposit:");
    Account acc = findAccount(accountNum);
    if (acc != null) {
        acc.deposit(amount);
    } else {
        System.out.println("Account not found");
    }
}
}

```



```

public void performWithdrawal () {
    System.out.println ("Enter Account Number:");
    int accNum = Scanner.nextLine();
    System.out.println ("Enter Amount to withdraw:");
    double amount = Scanner.nextDouble();
    Account acc = findAccount (accNum);
    if (acc != null) {
        acc.withdraw (amount);
    } else {
        System.out.println ("Account not found");
    }
}

```

```

public void showAccountDetails () {
    System.out.print ("Enter Account Number:");
    int accNum = Scanner.nextInt();
    Account acc = findAccount (accNum);
    if (acc != null) {
        acc.displayAccountDetails();
    } else {
        System.out.print ("Account not found");
    }
}

```

```

public void updateContact () {
    System.out.println ("Enter Account Number:");
    int accNum = Scanner.nextInt();
    Scanner.nextLine();
    Account acc = findAccount (accNum);
    if (acc != null) {
        System.out.println ("Enter new email:");
        String email = Scanner.nextLine();
    }
}

```



```

System.out.print("Enter Phone Number");
String Phone = Scanner.nextLine();
acc.updateContactDetails(email, Phone);
} else {
    System.out.println("Account not found");
}
}

public void mainMenu() {
    while (true) {
        System.out.println("\n -- Banking Application Menu");
        System.out.println("1. Create Account");
        System.out.println("2. Deposit");
        System.out.println("3. Withdraw");
        System.out.println("4. Show Account Details");
        System.out.println("5. Update Contact Details");
        System.out.println("6. Exit");
        System.out.print("Enter your choice:");
        int choice = Scanner.nextInt();
        switch (choice) {
            case 1:
                createAccount();
                break;
            case 2:
                performDeposit();
                break;
            case 3:
                performWithdrawal();
                break;
            case 4:
                showAccountDetails();
                break;
        }
    }
}

```


case 5:

update(contact C);

break;

case 6:

System.out.println("Exiting application. Goodbye");

return;

default:

System.out.println("Invalid choice. Please try again");

}

}

}

}

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